



Patricia Rodrigo Barrio

Biomedical Engineer | Artificial Intelligence | Applied R&D

Biomedical Engineer with an MSc in Data Analysis Engineering, Process Improvement and Decision-Making from the Polytechnic University of Valencia. My profile combines a strong background in biomedical engineering with advanced training in data analysis, artificial intelligence and computational modelling.

I have experience in applied research, clinical data analysis and the development of solutions for real-world biomedical problems, including data processing, metric generation and results validation. I am an analytical, curious and persistent person, quick to learn new tools and able to turn complex technical problems into clear, rigorous and reproducible solutions.

CONTACT

-  +34 619 32 63 85
-  patriroba03@gmail.com
-  Burgos, Spain
-  LinkedIn: Patricia Rodrigo Barrio
-  GitHub: prodbar
-  Portfolio: prodbar

EDUCATION

-  Polytechnic University of Valencia
MSc in Data Analysis Engineering, Process Improvement and Decision Making
Sep. 2025 - Jul. 2026
Average grade: 9.3/10
-  Polytechnic University of Valencia
BSc in Biomedical Engineering
Sep. 2021 - Jun. 2025
Average grade: 8.2/10 - Bachelor's Thesis with Honours

PROJECT PORTFOLIO

- Personal portfolio featuring a selection of projects in data analysis, machine learning, optimisation and computational modelling applied to biomedical, industrial and engineering problems.
- Includes methodology, results, visualisations and access to the project code on GitHub.

LANGUAGES

Spanish - Native
English - C1 (Cambridge CAE)

AWARDS

May 2021 | First prize

VI Virtual Agri-food, Agri-environmental and Forestry Olympiad of Castilla y Leon

EXPERIENCE

I2MB - Institute of Mechanical and Biomechanical Engineering

Oct. 2024 - Jul. 2026

Valencia, Valencian Community (Spain)

Clinical Data Analyst & Researcher

- Analysis of real-world clinical data, applied research and computational modelling in biomedical engineering projects.
- Generation of metrics, indicators and visualisations to support results interpretation and decision-making.
- Participation in the PREVILI project, focused on the analysis of hospital clinical data from neonatal and paediatric patients.
- Development of the BIOPINNs project, focused on physics-informed neural networks for hyperelastic biomechanical modelling, integrating deep learning, physical constraints and comparison against FEM.

Morcillas Rioseras Gourmet

Jul. 2024 - Sep. 2024

Burgos, Castile and Leon (Spain)

Production & Quality Control Support

- Recipient of the Campus Rural 2024 scholarship.
- Supported production, operational organisation and quality control activities in an industrial environment.
- Applied an analytical approach to real-world problems involving organisation, quality and process management.

TECHNICAL SKILLS

Programming & Data: Python · SQL · R · MATLAB

Machine Learning & Deep Learning: scikit-learn · TensorFlow/Keras · classification · regression · model validation

Biomedical Processing: image processing · signal processing · feature extraction

Visualisation & Optimisation: Power BI · Advanced Excel · OpenSolver